



Discovery

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Intro:

Several times a month, either I'm looking for something to watch or entertaining friends when someone says "let's watch something." Phones come out, and everyone starts Googling the same things ["good movies to watch", "action movies to watch", "drama movies that have some comedy", "good shows with short episodes"]. An hour later, we finally make a decision but nobody's happy.

The Problem:

Every month, more than 1.5 million people in the U.S. open up Google - not Netflix - to answer the question Netflix exists to answer: *What should I watch tonight?* The Google search "What to watch" draws 1.2 million U.S. searches a month, "good movies to watch" draws another 165,000-246,000 and there's a long tail running past 70,000 keyword variations. People also don't just search once. According to Google trends, the same viewer cycles through five or six rephrasing attempts – "feel good movies" → "best feel good movies" → "feel good netflix movies" → "good movies to watch" before landing on the movie for the night. Then comes the second search "where to watch _____" which throws users all over the place for movies. Netflix is barely in this Google interaction. On the results page for "what to watch," Netflix ranks 6th in traffic- capturing around 5% of the ~975,000 monthly visits to the top 20 sites. The #1 result belongs to IMDb owned by Amazon and the second being Rotten Tomatoes which pulls about 7.7x more visits than Netflix. Both of these along with 13 of the top 20 results are cross-service tools that send users to all sorts of streaming platforms and away from Netflix. **Netflix has the most sophisticated personalization engine in streaming and knows its catalog better than anyone; but it doesn't have a place for viewers to simply say what they're in the mood for — so night after night, they say it to Google instead.**

Insights:

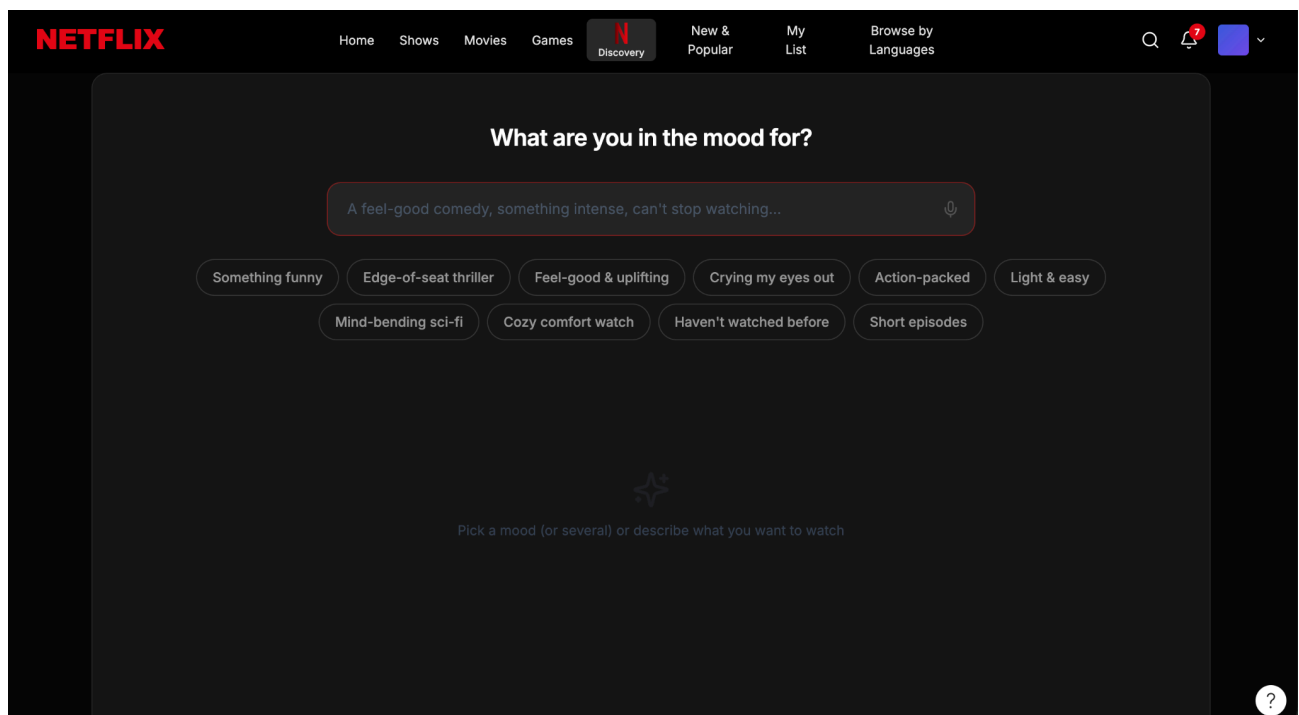
When Netflix's FM-Intent model introduced the prediction of four dimensions of intent (discovering or continuing, genre, movie or show, new or from the back catalog) it improved next-item accuracy by 7.4% over the strongest baseline they tested. The system reasons in terms of intent but has to reconstruct that intent from behavioral traces - what users have hovered over, added to lists, abandoned, time of day, or what they watched the week before. Netflix's own paper describes intent as "**often not directly observable** but crucial for providing relevant recommendations". In their June 2026 article, the GenPage team published a finding alluding to the same point. As they were building the generative model for the Netflix homepage

they tried to improve quality in two ways: Scaling the model or giving it better information about the request. The result was clear: scaling the model about 7.5x improved the loss metric by 1.3% and giving it better request context improved it 6.9%. They state “Personalization quality appears to be **bottlenecked first by the information and representation available to the model**, and only then by capacity”. **This is the gap. Netflix has built a system that reasons in terms of intent but ultimately guesses the intent.**

Let’s take a sentence both my family and I have Googled before: “New, sad drama movies to watch”. Referring to the four dimensions of intent that FM-Intent model predicts before giving a title, it’s evident that three of the four dimensions (genre, movie, new) are stated in under 7 words. Allowing users to state their intent is the missing input to Netflix’s insanely capable intelligence.

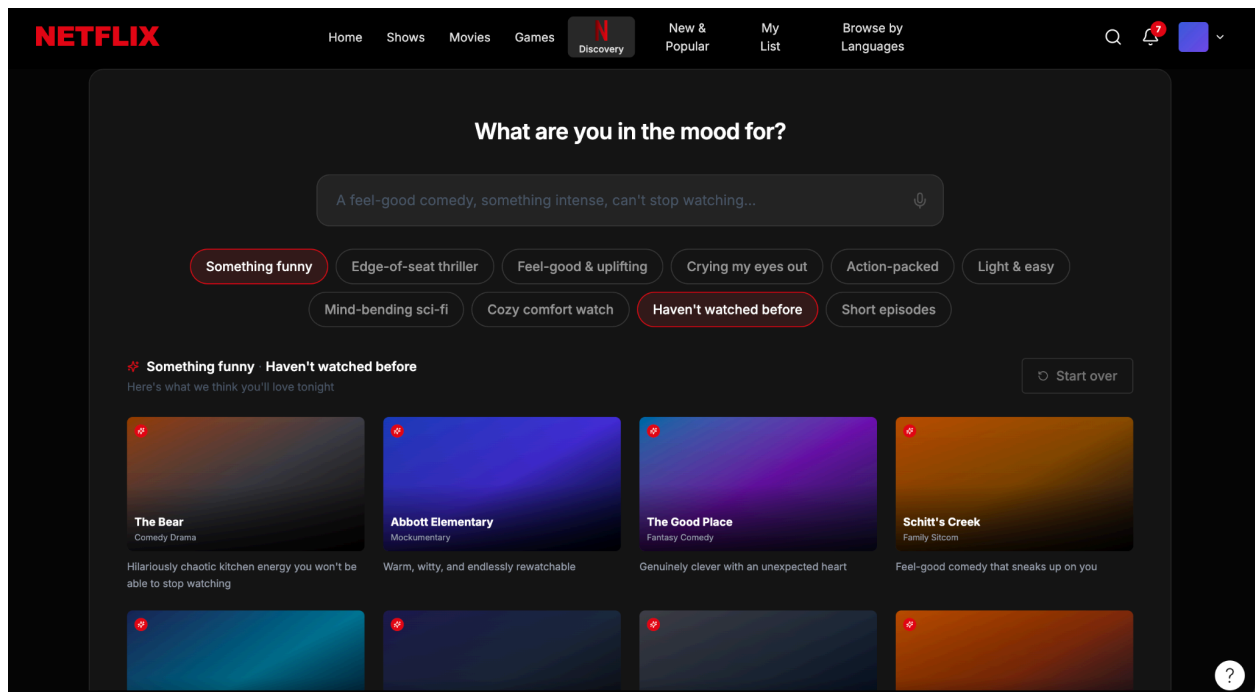
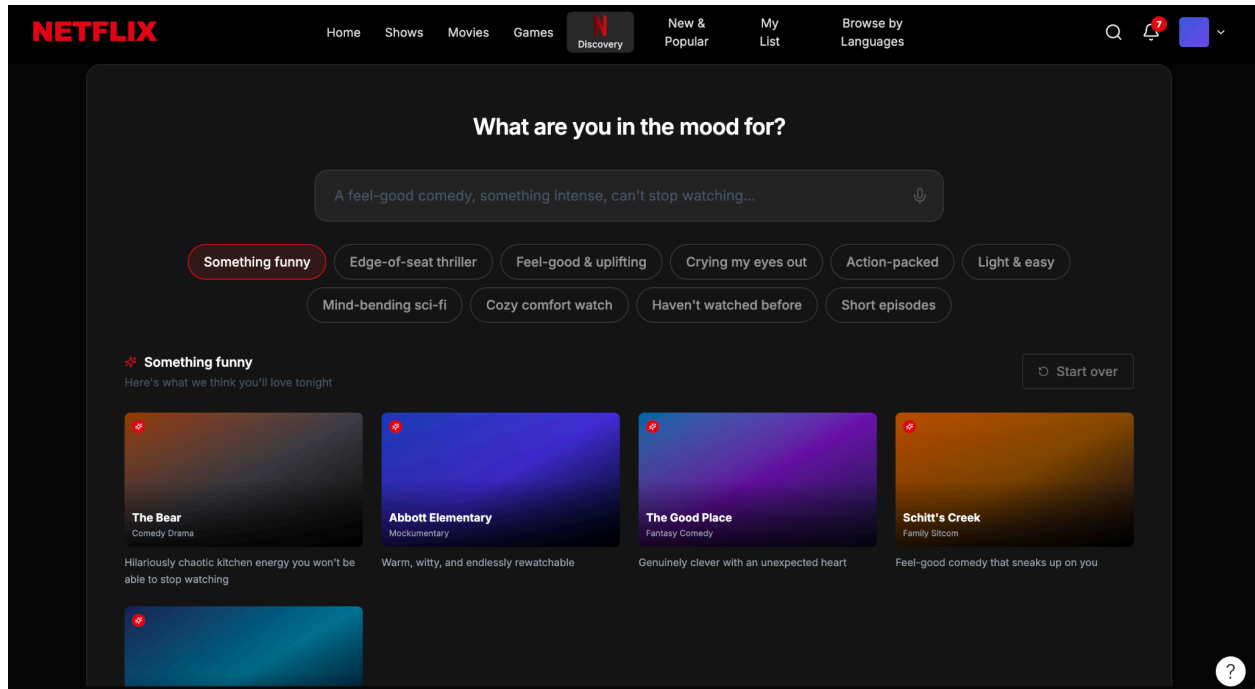
The Feature:

Netflix Discovery is an interactive input field with three ways for users to answer the question of intent (What are you in the mood for?) As shown below, users can either type it in a single free-text field, say it with a mic input, or tap it with the “Mood Chips”.



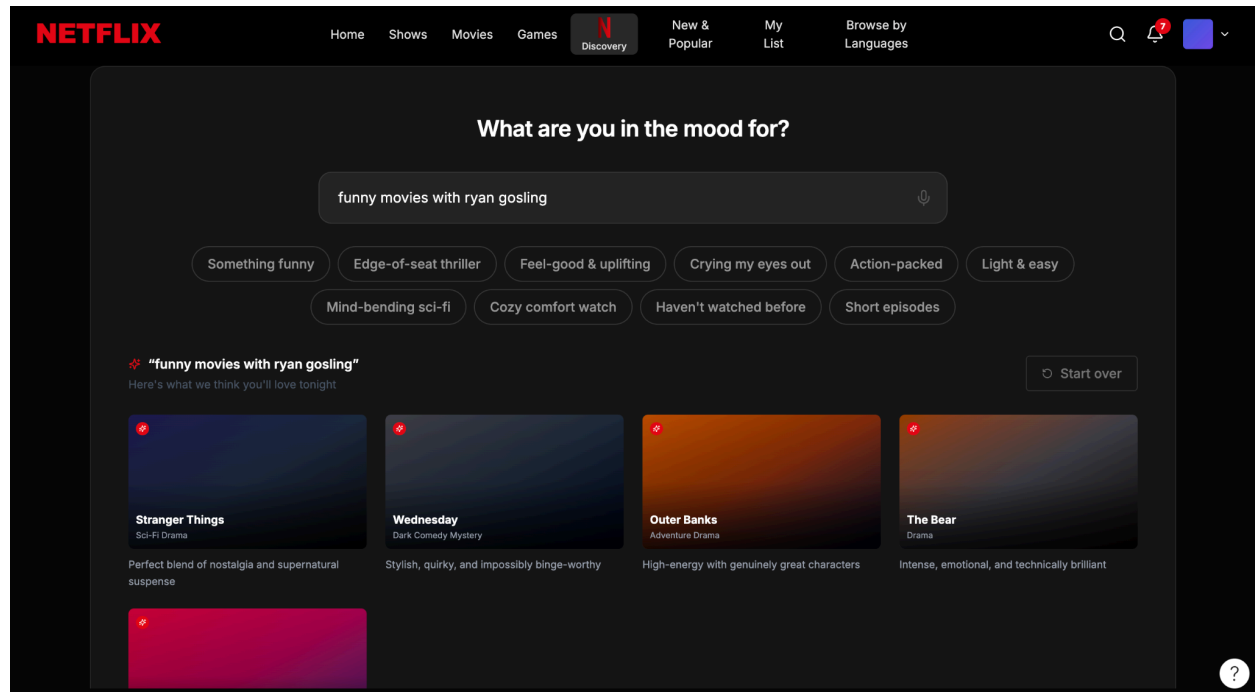
As the user taps on the Mood Chips, recommended titles show up in real time. The titles change as different mood chips are chosen and the intent is narrowed down. The mood chips build on each other (shown below) to reproduce a typical Google search query. A user starts with “Something funny” adds “Haven’t watched before” and then might add “Short episodes”. The result is:

“Something funny” + “Haven’t watched before” + “Short episodes” == the FM-intent model has three of the four predictors: New (to user), Genre, Show.



The feature accepts free text and the result is the same. As shown below, a user can type “Funny movies with Ryan Gosling” and the result is:

“Funny” + “Movie” + “Ryan Gosling” = 2 out of the 4 predictors (movie and genre) and adds the detail about the Ryan Gosling casting as additional help to narrow down.



The System Exists:

Netflix Discovery is a combination of Netflix's existing system architecture.

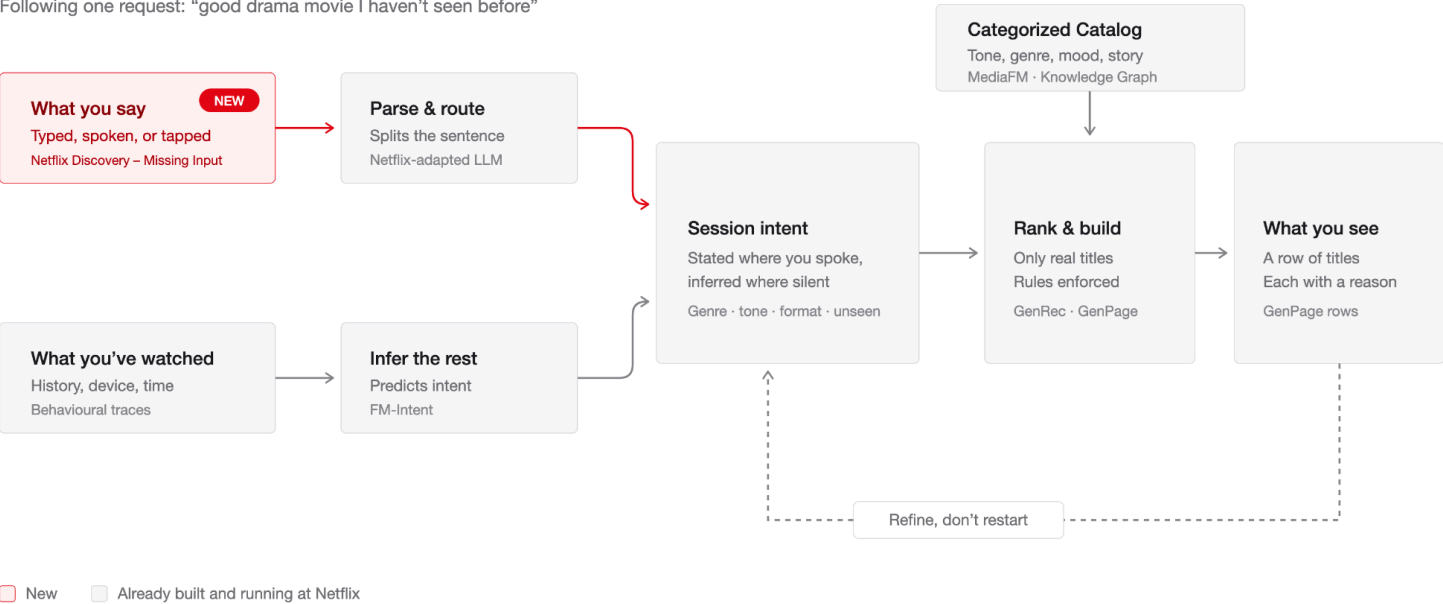
Categorization - Netflix's MediaFM tackles multi-label classification of hook clips into 100 tone categories. There are two important aspects to the feature: First, multi-label. A clip can be both creepy, scary, and humorous simultaneously (stated examples in articles). Second, the model reads "up to 512 shots at a time" so it understands each shot in the context of the ones around it. Since both of these capabilities are built, it would be possible to classify titles by their degree of tones or genre, therefore tackling inputs such as "a drama movie with some comedy" or "scary but not too scary show with action".

Real-Time Title Recommendation - GenPage "treats the user history and **request context** as the prompt, and autoregressively generates the entire homepage as the response". In this case the user's input, whether text (and/or voice to text) or mood chip selection, is the request context. The second part stays the same. The page is then generated autoregressively with each token conditioned on everything already on the page allowing for full-page optimization and no duplication. (Known Gap - The input for the prompt and response system is currently built as domain-specific tokens, not text. I expand on this in the Design Spec)

Intent Engine - FM-Intent is the driving force behind the recommended titles. FM-intent allows for predicted user intent to be split into four predictive signals. The stated intent would override the predicted intent on the dimensions the user specified but FM-Intent would continue to supply the prediction on the dimensions users did not touch, **creating a hybrid intent model** using user input.

One new input. Everything else already exists.

Following one request: "good drama movie I haven't seen before"



What success looks like:

Netflix Discovery is a new feature and can be evaluated in an “ordinary” way. Ship Discovery to one group of users and compare the outcomes. Discovery, however, is an opt-in feature - a tab users would choose to open meaning that results should be reported **intent-to-treat** where all users assigned to the Discovery group (not just the ones who used Discovery) are compared against all users in the control group (where Netflix Discovery is not available). This would enable us to measure improvements Netflix would achieve by shipping the feature instead of the difference between users who used Discovery vs users in general (Intent to treat method of measurement is expanded on in the Technical Spec).

The metrics below would be used for measuring success:

Metric	Existing/Proposed	Explanation
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Core engagement metric used for launch decision	Existing	Netflix's internal quality bar for launches. The GenPage articles states GenPage feature was judged on it vs. a production baseline.
Sessions ending without playback	Existing	In the article <i>Measuring Impact of Personalized Recommendations</i> it's stated that "watching nothing" is an explicit outcome rather than absence of data. This would allow us to track the explicit outcome and treat it as an "outside option."
Abandonment rate	Existing	Tracked by GenPage and gets assigned a negative reward. We can track a positive change due to the titles chosen through this feature. Reported with take fraction to measure honest persuasion, not clickbait.
Take fraction on rationale lines	Proposed	Of users who see a title's one-line reason (or synopsis), how many choose to watch. Netflix built and A/B validated this metric in their <i>Evaluating Netflix Show Synopses with LLM As-A-Judge</i> article.
Refinement turns in a session	Proposed	Google trend research shows a Google search for a title takes 5 to 6 reformulations. Netflix can target <i>under 3</i> to show the search loop wasn't just moved from Google to Netflix but reduced.
Time from session start to play	Proposed	Direct measurement of the problem Discovery exists to solve. Can directly compare with time users spent looking for a title without Discovery.

Success relies on making sure the goals and rules within Netflix are followed. Thus, the following guardrails should be put in place.

Consumption Diversity: According to Netflix's metrics, increased concentration is harmful. The Netflix Discovery feature should not narrow the viewing spread across the Netflix catalog. This metric must be tracked during testing to ensure the feature steers away from funneling viewers to the same titles.

Mid-Catalog Growth: Netflix states in their article *Measuring the impact of Personalized Recommendations* that recommendation's effect is split into Exposure, Selection, and Targeting. Netflix Discovery touches on all three but its main intent is to produce a better title choice than the recommendation system would've given without the user prompt This is by Netflix's definition, targeting. The article states "Targeting is also not uniform across the catalog. It is strongest for mid-popularity titles: titles with enough potential audience to be broadly relevant, but not so much mass appeal that everyone already knows about them. Very popular titles need

less help from targeting, while very niche titles may have too small an audience to target effectively. The largest gains come from the middle of the catalog, where personalization helps connect specific titles with specific member tastes". Therefore, success looks like the mid-catalog share increasing.

****It's possible that most of the benefit with Netflix Discovery is convenience:** Serving users who were already going to watch and therefore would be "Selection" as Netflix defines it. Selection and Targeting are distinguishable, however, as selection won't shift the distribution of what gets watched while targeting should raise viewing among mid-catalog titles.

What I don't know:

Search Data: The research showed that this problem is real and widespread. However, it does not establish how much it costs Netflix. Searches don't mean a user is leaving Netflix to watch elsewhere, they demonstrate unmet intent, not churn.

Questions like [How long does a browsing session run before playback? What share of sessions end without any playback? Does the length of a browsing session correlate to user retention?] can not be answered without numbers Netflix holds.

System Design: There are three parts of the system architecture that I inferred upon and don't know if they exist or how they work.

Media FM classifies clips across ~100 categories but there is no mention of classification by "degree." Searches like "scary but not too scary" rely upon a gradient of tone. Additionally, from my research, Media FM classifies clips across ~100 categories but ends up attributing just one of ~10 genres to a title in general, therefore the Discover feature may struggle to make tone connection at a title level. The tone classification could be moved to titles. (I briefly expand on this in the Design Spec)

GenPage cannot read a sentence. It takes in prompt tokens so parsing a sentence into a token GenPage would be able to read requires parsing handled by a separate model and arriving as structured intent. Nothing in Netflix's published work describes handling *stated* intent, the entire infrastructure is built to infer intent.

Measurement: My proposal defers to Netflix's core engagement metrics used for launch decisions. This leads to questions around how engagement is measured and new feature engagement benchmarks.

There is no publicly available research/data around the cost of a wrong answer for Netflix. A row misaligned on user intent is forgettable, but risking a bad answer to a direct question may cause increased frustration and push users off the platform.

It is also difficult to calculate whether the Netflix catalog has the depth to answer specific questions with niche tones. A recommended row made up of 3 to 4 weak links to the user input is worse than a good browse row with no user input. There would have to be analysis done on the potential combinations of moods users can input and the percentage of confidence (alignment to input) in the titles Netflix returns.

Sources:

The bulk of my research was done using articles posted on the *Netflix Technology Blog*. I specifically focused on 11 articles which touched on personalization, recommendations, categorization, GenPage, and how to measure the impacts of all four. I researched search data using Google Trends, Google Keyword Search (using a Google Ads Account), SemRUSH (paid SEO tool), Ubersuggest, and Clicks.so. All the articles I read are linked in the GitHub repo below. There are summarized versions of each one and a preliminary synthesis created to try and define the architecture design at Netflix. In addition, the screenshots I took of the keyword and search data are also linked in the “screenshots” folder in the repo.

<https://github.com/orozcotorres-samuel/netflix-discovery-proposal>

Thank you for taking the time to read this!

I created an expanded spec on certain topics below. I'd love to know your thoughts!



Expanded Design Spec

Categorizing by Degree:

MediaFM works at the shot level - the segment between two camera cuts. For each shot it reads video, audio, and dialogue, then it reads it in context looking at up to 512 surrounding shots from the same title. It produces a *position*- a 2,304 dimension embedding. A small classifier trained on top of those frozen embeddings assigns a specific tone out of a 100 category taxonomy maintained by Netflix's internal Metadata & Ratings team. The classifier uses multi-label categorization, which means that a shot can register as multiple tones such as "creepy" and "humorous" at the same time. The result of this is tone measurement for every shot within a title, but not for the entire title itself.

A user asking for "a drama with some comedy" is asking a title level question. Titles are currently given one of eleven predetermined genres, there's no flexibility and no room for degree. Title level tone should result from the prevalence of shot level tones. For each tone category, Netflix would measure how much that tone shows up and attach percentage values to it. For example, a film where 33% of the runtime reads humorous tone is considered a comedy. A film where 5% of the runtime reads humorous and 25% of the film reads drama is considered a drama with some comedy. These are not different measurements. They are the same measurement with thresholds attached. The vocab a user might use in their stated intent ("some", "really", "a bit of") maps onto the different tone thresholds.

With this, Netflix would be producing intensity from a system that doesn't have a notion of intensity.

Open questions:

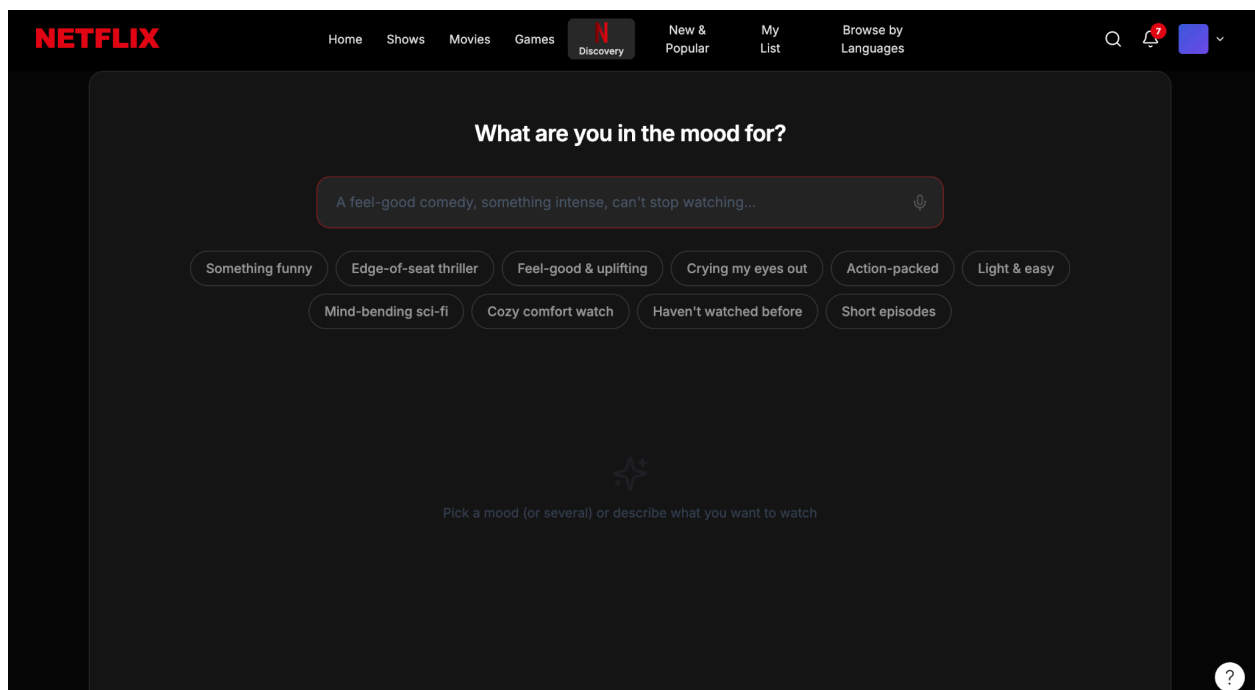
- Difference between prevalence of a certain tone and the title genre. Would the categorization of shots with a measure of their prevalence help improve the genre given to a movie?

- Different tones aggregate differently. For example, a user asks for a movie that is "scary but not too scary". Horror movies usually contain a dramatic or creepy tone for 95% of the movie and there are massive spikes on jumpscare that can lead to a "scary" tone. If 5% of the movie is genuinely terrifying it might seem to be the perfect movie to fall under the "scary but not too scary" category but, in reality, it's above the user's "scary" threshold. It'd be good to address

whether some tones are prevalence driven and others (like horror or violent) are peak driven and therefore we can establish a cap and address the question of “scary but not too scary”.

Handling Stated Intent:

In the “FM-Intent” article, Netflix states that their most powerful LLM model, FM-intent, has four dimensions: *action_type* (user is discovering something new or continuing something), *genre_preference*, *format* (movie or show), *time_since_release* (new, recent, evergreen catalog). These dimensions provide the parse target for the stated user input. User input is given as text (derived by typing or voice to text) or the mood chip selection.



Mood chips:

A mood chip at its most basic is a pre-filled version of one of the four dimensions of intent.

“mind bending sci-fi” → (genre == sci-fi)

“edge of my seat thriller” → (genre == thriller)

“Haven't watched before” → (action_type == discovery)

“Short episodes” → (format == movie)

“Nostalgic watch” → (time_since_release == evergreen catalog)

The recommended titles shown after clicking one mood chip is roughly the same information content of a recommended row on the home page. The mood chips value grows as multiple mood chips are selected and the 4 dimensions of intent are represented.

“Something funny” + “Short episodes” + “New” == 3 of the 4 dimensions of intent. The efficacy of mood chips is that they all come **pre-mapped to a dimension. The need to parse the stated intent is gone.** For example “short episodes” will always map to [format == movie], “new” will map to [time_since_release == new] and genre mood chips will map directly to their chosen genre.

Mood chips are also important because **they are the on-ramp to free text searches or primers.** Priming occurs when you expose users to a specific set of words or categories that influence how they respond to the open text fields. In adding mood chips, you are also teaching users how best to input their specific mood.

Free Text:

The most difficult open question about this proposal is the parsing of the free text stated intent. Parsing is expensive and requires extensive LLM training. The two options below represent the main two ways to do this:

The definition of parsing I chose to work with is “analyzing an input against a grammar and building a structure representation out of its constituent parts”

Option 1. Match against a fixed vocabulary. The cheapest way does not involve parsing at all but instead matching. Netflix would build a library of curated mood chips, each one a specific intent object. If the library of mood chips already exists then the stated intent does not need to be parsed, it can be matched. The system would take the stated input and compare it against the mood chip set, taking (matching) whichever chips clear a predetermined similarity threshold and union those chips to be the resulting input token which GenPage is built to handle.

Pros: The mood chip library and its possible combinations is finite, meaning that every downstream result can be precomputed and refined before a user even inputs their intent. This would also trickle down to synopsis and rationale lines (if included). Each synopsis and rationale line can be predetermined to match the chosen combination of mood chips. For example, if an original title synopsis doesn’t include something about the title being a comedy but there is a comedic tone to it, a new title synopsis talking about the comedic tone could show if the comedic mood chip were to be chosen.

Cons: Lack of degree flexibility. If we take the input “sad movie but not too sad with some comedy” matching to mood chips would likely ping “sad” + “comedy” + “movie”. It would most likely ignore “but not too sad” and “with some” input unless the mood chips with degrees of tone were created from the beginning. The lack of degree flexibility narrows the scope and strength of the recommendations.

Parse in the open. Rather than treating the parsed object as an internal artifact, the Discovery page could show it back to the user. The parsed slots become visible mood chips the

user could add, remove, or adjust before the results resolve. To continue the example, “sad movie but not too sad with some comedy”, the parser will see “Movie” + “Sad” + “Comedy” and will be given the ability to click any of the chosen moods off.

Option 2. Slots with a residual. This approach splits into two parts, one model pass reads what the user stated and writes a record of it structured into the four dimensions FM-Intent uses. As the model writes the record, constrained decoding would ensure it sticks only to acceptable values (for example “horror” would fall under one of the 11 genres Netflix classifies, “dystopian” might not) and will not output anything outside the acceptable list. Fields the member didn’t mention are left empty and FM-Intent’s prediction kicks in.

The second part deals with all the information contained in the state input that did not fit into the four dimensions of FM-Intent. For example “sad movie with comedy my wife would like”, the “my wife would like” part can not be parsed into one of the four dimensions. These sections would not be discarded; they would be kept as a “residual” that wouldn’t act as a filter for titles but would nudge the final title recommendations slightly. The argument for keeping the residual information comes from the GenPage article where it states that enriching the prompt gave a ~6.9% improvement in loss.

Pros: Out of the two options, this one does not discard the residual part of the stated intent leading to an enriched prompt fed to GenPage. The output is guaranteed to be well formed (like the matching option).

Cons: There is routing ambiguity with tone labels. Input that touches on multiple tones might route inconsistently, falling into the structured output one input and residuals the next.

The matching option is the most cost effective and integrates easiest with the existing system design, it falls short on the extra details that could enrich the prompt but can not be matched. You could lessen the gap by expanding the existing mood chip library, but it will always miss. The second option captures the details but there is ambiguity on how it would handle them. If I had to choose between these two options I would go with option one and really refine the library of mood chips that would be matched against. It would be the quickest, allowing for shorter latency from input to recommendations and the most cost effective, making it the best option besides creating a custom LLM parser focused solely on deducing user input (taking residuals into the delivered output as well).

Measuring Impact:

Discovery is a net new feature that can be evaluated the “ordinary” way: ship it to one group and withhold from another, compare the outcomes. It is important to note, however, that Discovery is opt-in. It’s a tab that users can choose to open, so the test group contains those who used the feature and users who did not. The results should therefore be reported as intent-to-treat. Every member in the group that gets Discovery should be compared against every member in the control group whether or not they used Discovery.

Members who used the Discovery feature are self-selected- those who couldn't find something to watch, browse the most, or are the most willing to try something new. Comparing this group to the general population measures the difference between two kinds of people and attributes it to the feature.

Intent-to-treat (ITT) is unbiased, but it dilutes the effects since non-users contribute null results to the average. An example is shown below.

Adoption in Discovery Group	Measured Effect Among Users	Reported ITT effect
3%	25%	0.75%
8%	25%	2.0%
20%	25%	5.0%
8%	10%	0.8%

With ITT it is evident that **adoption is a multiplier on every result the feature can produce**. Adoption is very important with ITT, if a feature works extremely well for the people who find it but it's not found by many people the effect may not show clear significance.

ITT answers the question "what does Netflix get by shipping?" It does not tell you whether the feature is effective. A high effect among users (signifying that the feature works well) paired with a low adoption rate will pin the ITT measured effect at a low number which can lead to scenarios like "the feature doesn't work" or "the feature works but no one found it." This is why it's important to measure other metrics such as: Sessions ending without playback, abandonment rate, and take fractions on synopsis or rational lines.